Riphah School of Computing & Innovation
FYP Framework v1.0

Final Year Project Types for BS Computing Programs

The following Final Year Project / Capstone Project categories provide students with multiple pathways aligned with their career aspirations, including software development, research, entrepreneurship, industrial collaboration, consulting, and emerging technologies. Each category has distinct objectives, outcomes, and deliverables while emphasizing problem-solving, innovation, professionalism, and effective communication.

Project Type	Suitable Programs	Description	Expected Outcomes	Expected Deliverables	AI Usage
Product Development Project	CS SE IT	A project focused on designing, developing, testing, and deploying software or hardware solutions to solve a real-world problem. Students follow the complete Software Development Life Cycle (SDLC) and apply software engineering principles.	• Demonstrate software engineering competencies. • Develop a reliable and usable computing solution. • Apply modern development tools and technologies. • Validate the solution through testing and evaluation.	• Software application or computing system • Source code repository • Software Requirements Specification (SRS) • System Design Document (SDD) • Test Plan and Test Report • User Manual • Technical Documentation • Final Project Report	• No AI usage/assistance in core functionalities and documentation • AI application through APIs is allowed
Entrepreneurial Project	CS SE IT	A project that develops an innovative technology-based product or service with commercial potential. Students identify market needs, validate business opportunities, and build a Minimum Viable Product (MVP). Student can extensively use AI.	• Develop entrepreneurial and innovation skills. • Validate market demand. • Create a scalable business model. • Demonstrate product feasibility and commercialization potential.	• Minimum Viable Product (MVP) • Business Plan • Business Model Canvas • Market Research Report • Financial Feasibility Analysis • Pitch Deck • Product Demonstration	Students can extensively use AI for product development and documentation.

Continuation Project (Existing product based Project)	CS SE IT CY DS AI CGV	A project carried out in collaboration with an industry partner to solve an actual organizational problem using industrial standards, methodologies, and professional practices.	• Deliver practical business value. • Gain industrial experience. • Meet client expectations and professional quality standards. • Improve teamwork and stakeholder communication.	• Industry-ready solution • Client Requirements Document • Deployment Plan • Client Acceptance Report • User Documentation • Final Project Report	• AI usage allowed depending on the project for new features development. • AI application through APIs is allowed • No AI usage in documentation.
Research Project	CS AI CY DS	A project involving scientific investigation of a computing problem through literature review, experimentation, data analysis, and validation to generate new knowledge or improve existing methods.	• Apply scientific research methodology. • Evaluate existing techniques. • Develop and validate a novel approach. • Produce publishable research findings.	• Literature Review • Research Proposal • Experimental Design • Dataset (if applicable) • Prototype (optional) • Research Paper Draft • Final Research Report	• No AI usage/assistance in core functionalities and documentation • AI application through APIs is allowed
IoT & Embedded Systems Project	CS SE IT	A project involving the design and implementation of intelligent systems integrating sensors, embedded devices, communication technologies, cloud platforms, and software applications.	• Develop integrated hardware-software solutions. • Demonstrate IoT architecture design. • Collect and process sensor data. • Validate system performance and reliability.	• Hardware Prototype • Embedded Software/Firmware • Cloud Dashboard • Mobile/Web Interface • Circuit Diagrams • Technical Documentation	• No AI usage/assistance in core functionalities and documentation • AI application through APIs is allowed
Case Study Project	IT CY DS	A systematic investigation of an existing organization, technology, information system, or computing problem to analyze current practices,	• Develop analytical and critical thinking skills. • Evaluate existing systems objectively. • Recommend evidence-based solutions.	• Case Study Report • Current State Analysis • Root Cause Analysis • Recommendations • Implementation Roadmap	• Limited AI usage/assistance in core functionalities and No Usage in documentation

		identify issues, and recommend improvements.	• Demonstrate professional reporting.	• Presentation	•
Consulting Project (AI/Data Analytics)	AI DS	A consultancy-based project in which students analyze organizational data and provide AI- or analytics-driven recommendations to improve business performance and decision-making.	• Apply AI and data analytics to real organizational problems. • Generate actionable business insights. • Improve decision-making using data. • Demonstrate consulting and communication skills.	• Business Problem Analysis • Data Analysis Report • AI/ML Models or Analytics Dashboard • Business Recommendations • Executive Presentation • Consulting Report	• No AI usage/assistance in documentation •
Consulting Project (Cybersecurity)	CY IT	A cybersecurity consultancy project involving security assessment, vulnerability analysis, risk management, compliance evaluation, and security improvement recommendations for an organization.	• Identify security vulnerabilities. • Assess organizational cyber risks. • Recommend appropriate security controls. • Demonstrate cybersecurity consulting practices.	• Security Assessment Report • Vulnerability Assessment/Penetration Testing Report (where applicable) • Risk Assessment Matrix • Security Improvement PlanExecutive Summary • Consulting Report	• No AI usage/assistance in documentation •
Consulting Project (Cloud & DevOps)	IT CS SE	A consultancy project focused on evaluating cloud infrastructure, DevOps maturity, application modernization, automation, and cloud adoption strategies for an organization.	Assess cloud readiness. Recommend scalable cloud architectures. Improve deployment automation. Optimize operational efficiency.	Cloud Readiness Assessment Cloud Architecture Design DevOps Maturity Assessment CI/CD Pipeline Design Migration Strategy Consulting Report	• No AI usage/assistance in documentation
Consulting Project (Digital Transformation)	IT CS SE	A project that analyzes organizational processes and develops a comprehensive digital transformation strategy using modern computing technologies such as AI, cloud computing, automation,	Evaluate organizational digital maturity. Recommend technology-driven transformation initiatives.	Digital Maturity Assessment Business Process Analysis Technology Roadmap Digital Transformation Strategy Cost-Benefit Analysis Executive Presentation	• No AI usage/assistance in documentation

		enterprise systems, and data analytics.	Improve operational efficiency and customer experience. Develop a strategic digital transformation roadmap.	Consulting Report	

Common Deliverables for All Final year Projects

Regardless of the project category, every Final Year Project should include the following core deliverables:

1. Project Proposal
2. Problem Statement and Objectives
3. Literature Review / Background Study
4. Project Planning and Timeline
5. Requirements Analysis (where applicable)
6. Design and Methodology
7. Implementation or Investigation
8. Validation, Testing, or Evaluation
9. Final Project Report
10. Poster and Demonstration
11. Oral Defense Presentation
12. Source Code, Prototype, Dataset, or Supporting Artifacts (as applicable)
13. User and Technical Documentation
14. Reflection on Project Outcomes and Future Work
15. 9 hours per week must be spent on the project for a complete project of 300 hours.

Learning Objectives Across All Final Year Types

All Final Year project categories are intended to enable students to:

- Apply computing knowledge to solve complex real-world problems.
- Demonstrate technical competence in their chosen specialization.
- Integrate theoretical concepts with practical implementation.
- Conduct independent inquiry, analysis, and critical evaluation.
- Work effectively as individuals or within multidisciplinary teams.
- Communicate technical ideas professionally through written reports, presentations, and demonstrations.
- Follow ethical, legal, and professional standards in computing practice.
- Deliver solutions that create measurable academic, industrial, commercial, or societal impact.